

15.5 Answers

① a) $m = \frac{4}{3}$, $(\frac{4}{3}, 0)$

b) $m = \frac{27}{2}$, $(\frac{8}{5}, \frac{1}{2})$

② $(\frac{3}{8}, \frac{3\pi}{16})$

③ $I_x = \frac{189}{20}$, $I_y = \frac{1269}{28}$, $I_0 = \frac{1917}{35}$

④ $I_x = \frac{4}{9} \rho$, $I_y = \rho(\pi^2 - 4)$

$$\bar{\bar{x}} = \sqrt{\frac{\pi^2 - 4}{2}}, \quad \bar{\bar{y}} = \frac{\sqrt{2}}{3}$$